W3 Lab B: Sound Design and Dynamic Audio

COMSM0127 Immersive Interaction and Audio Design · Wed 18 November · Full worksheet on Blackboard: **W3 Lab B**

Your task. Rebuild the mystery sound two ways, and make it change. Everyone does both parts. At 09:50 everyone stops for five minutes, then swaps at 09:55. **Recording outside? Back in the room by 09:35 (first half) or 10:10 (second half).**

	09:25 – 09:50	09:55 – 10:20	10:25
Groups 1, 2	**Part A: Recorded**	Part B: Procedural	Everyone back together: listen
Groups 3, 4	**Part B: Procedural**	Part A: Recorded	

First: listen

Headphones on, eyes closed. As a group: how many separate parts can you hear? High or low? Constant or in bursts? Give each part your own name. One part per layer.

Our parts:

Open the project

1. **github.com/RichSF1210/W3B_FireLab**: green **Code** button, **Download ZIP**.
2. **Unblock it first:** right-click the zip, **Properties**, tick **Unblock**, OK. Then right-click, **Extract all** (Desktop is fine). Skip this and Windows blocks the extract.
3. In File Explorer, `Assets/W3B_Fire/Scenes`: double-click **W3B_Student_Shell**. Unity opens the project (a few minutes the first time). No need for Unity Hub.
4. Two silent campfires: **Campfire A (Recorded)** for Part A, **Campfire B (Procedural)** for Part B. Each has three empty layers: one part of the sound per layer. `W3B_Complete_Reference` has both finished, for later.

Part A: Recorded (on Campfire A)

1. Take a Zoom recorder outside. Record **several takes** of each part. Get close; about 10 seconds each.
2. Files off the Zoom: USB cable (choose *Card Reader*) or the SD card. Trim in Audacity (Ctrl+T). Short sounds: one per file, at least four. Constant sounds: a few seconds that loop. Export WAV, no spaces in names, into `Assets/MyAudio`.
3. **Short parts:** select a layer, **Add Component > MyRandomPlayer**. Lock the Inspector (padlock), drag your files onto **Clips**, unlock.
4. **Constant parts:** select a layer, **Add Component > SampleLoopLayer**, drag your loop into the Audio Source's **AudioClip**.
5. Press Play. What gives it away as a recording?

Your one line: make it vary

Double-click the Script field on My Random Player. Under `TODO`, remove the `//` so the line reads:

```
source.pitch = Random.Range(0.85f, 1.2f);
```

Save. Wait for Unity to finish, press Play. *The files never change; the playback never repeats.*

Part B: Procedural (on Campfire B)

Where the shape comes from: source (noise) → filter (what it seems made of) → envelope (how it moves in time: a snap is an instant attack and a short decay) → randomness (it never repeats).

1. Select a layer, **Add Component > NoiseLayer** or **BurstLayer**. Some parts need one, some the other.
2. Press Play. Move the sliders **while you listen**. Mute the other layers (Level 0) while you tune one.
3. Match each layer to a part on your list. **Write your best values down:** they reset when you stop.

NoiseLayer (a constant part)	BurstLayer (short random events)
Noise through a filter, with a slow wobble. Filter · Cutoff · Wobble · Level	Very short bursts at random moments. Bursts per second · Length · Brightness · Variation · Level

Your one line: make it respond

1. Select **Campfire B**, **Add Component > MyDynamicLink**.
2. Drag **Campfire B** into **Fire**, and your layers into **Burst Layer** and **Noise Layer**.
3. Double-click its Script field. Under `TODO`, remove the `//`:

```
burstLayer.burstsPerSecond = Mathf.Lerp(1f, 40f, i);
```

Save, press Play, select Campfire B and drag **Intensity** from 0 to 1. What changes? Is it more than volume?
Lerp goes from 1 to 40 as Intensity goes from 0 to 1: real-time parameter control (RTPC).

Stuck?

No sound	Level above 0? Audio Source volume up? Object ticked?
Red error	Check the line ends with `;` and the spelling matches exactly. Double-click the error to jump to it.
Changes vanished	You made them while playing. Make them again when stopped.
Out of time	Use the backup recordings in `Audio/Fire_Recorded`.

Listening safety: start with headphones low. Bursts can be sudden. Ask before recording people.

At 10:25, be ready to tell us

1. What parts did you hear? What did you call them?
2. Which part worked better recorded, and which procedural?
3. What did each line of code change?

Notes: